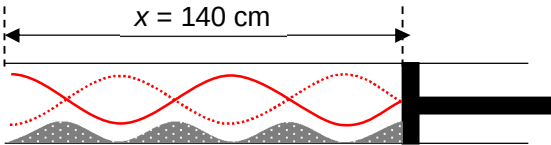
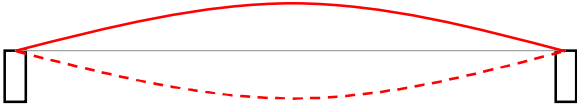
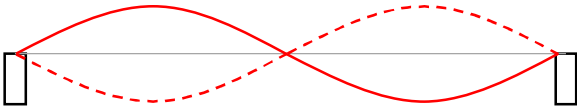
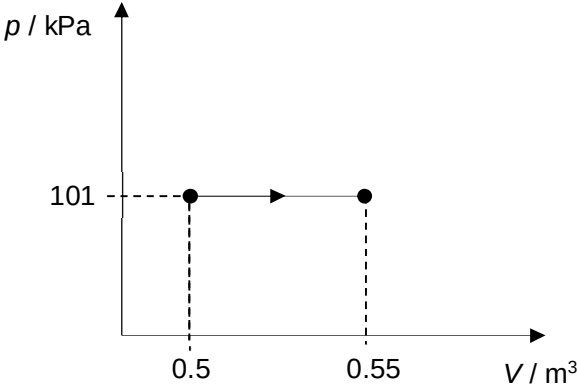


	is higher	
8(a)(i)	when two or more waves of the same type meet and overlap, resultant displacement at any point is the vector sum of the individual displacements	B1
8(a)(ii)	(the two sources have a) constant phase difference.	B1
8(b)(i)	progressive sound wave produced by the speaker <u>reflects</u> at piston <u>overlaps with incoming incident wave</u>	B1
	two waves are same type, have <u>same speed and frequency</u> , so forms <u>a stationary wave</u> with nodes and antinodes.	B1
	<u>piles of powder gather at the nodes (regions of no displacement) as the air particles remain stationary at the nodes.</u>	B1

Qns	Answer	Marks
8(b)(ii)	 $x = 1\frac{3}{4}\lambda = \frac{7}{4}\lambda$ $\left(\lambda = \frac{4x}{7} = \frac{4(140 \times 10^{-2})}{7} = 0.80 \text{ m} \right)$ $v = f\lambda = f\left(\frac{4x}{7}\right) = (400)\left(\frac{4(140 \times 10^{-2})}{7}\right) = 320 \text{ m s}^{-1}$	C1 A1
8(c)(i)		B1
8(c)(ii)	$v = f\lambda = f\left(\frac{1.5}{0.5}\right) = (150)\left(\frac{1.5}{0.5}\right) = 450 \text{ m s}^{-1}$	B1
8(c)(ii)	 $v = f\lambda =$ $f = \frac{v}{\lambda} = \frac{450}{1.5} = 300 \text{ Hz}$	B1
8(d)(i)	avg (translational) ke of particles is directly proportional to thermodynamic temperature of system in kelvin higher temperature so higher speeds, hence larger momentum change per particle per collision pressure is (normal) force per unit area. for same force, given larger change in momentum per collision time interval between successive collisions must increase so particles travel longer distances in between collisions (volume has be larger)	B1 (link ΔT to $\uparrow \text{KE}$, $\uparrow v$ to Δp) B1 (link pressure to force, $\uparrow \Delta t$ in $\frac{dp}{dt}$) any missing, max 1

Qns	Answer	Marks
8(d)(ii)	 $pV = NkT$ $\left. \frac{p}{Nk} \right _{\text{constant}} = \frac{T}{V}$ $\frac{T_{\text{initial}}}{V_{\text{initial}}} = \frac{T_{\text{final}}}{V_{\text{final}}}$ $V_{\text{final}} = \frac{V_{\text{initial}}}{T_{\text{initial}}} T_{\text{final}} = \frac{0.50}{27 + 273.15} (57 + 273.15)$ $= 0.55 \text{ m}^3$	B1 (flat line) B1 (all 3 labels)
8(d)(iii)	monoatomic ideal gas so $U = \frac{3}{2} NkT = \frac{3}{2} pV$ $\left(Nk = \frac{pV}{T} = \frac{(101 \times 10^3)(0.50)}{27 + 273.15} = 168 \text{ J K}^{-1} \right)$ $(N = 1.22 \times 10^{25})$ $(n = 20.2 \text{ mol})$ $\Delta U = \frac{3}{2} (p_{\text{final}} V_{\text{final}} - p_{\text{initial}} V_{\text{initial}}) = \frac{3p}{2} (V_{\text{final}} - V_{\text{initial}})$ $= \frac{3(101 \times 10^3)}{2} (0.55 - 0.50)$ $= 7575 \text{ J}$	C1 (Nk or N or n) C1 $\Delta \left(\frac{3}{2} pV \right)$ A1 (accept 2 s.f.)
8(d)(iv)	$pV = NkT$ $pV = \frac{1}{3} Nm \langle c^2 \rangle$ $\frac{1}{3} Nm \langle c^2 \rangle = NkT$ $\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$	B1

Qns	Answer	Marks
8(d)(v)	mass of 1 atom $m = \frac{M_r}{N_A} = \frac{84.0}{6.02 \times 10^{23}} = 1.4 \times 10^{-22} \text{ g} = 1.4 \times 10^{-25} \text{ kg}$ $\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$ $\langle c \rangle = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23})(57 + 273.15)}{\frac{84.0}{6.02 \times 10^{23}} \times 10^{-3}}}$ $= 313 \text{ m s}^{-1}$ or $\frac{1}{2} M_r \langle c^2 \rangle = \frac{3}{2} RT$ $\langle c \rangle = \sqrt{\frac{3RT}{M_r}} = \sqrt{\frac{3(8.31)(57 + 273.15)}{84.0 \times 10^{-3}}}$ $= 313 \text{ m s}^{-1}$	C1 (mass of 1 Kr atom) A1
9(a)(i)	spreading of waves into region of geometric shadow as they pass	P1